P measured, low noise

P + E measured, low noise

P + E measured, high noise

A

B

C

Concentration

time

M (True simulation)
E (True simulation)
P (True simulation)
E (Measurement)
P (Measurement)
M (Fit, known)
E (Fit, known)
P (Fit, known)
M (Fit, param)
E (Fit, param)
P (Fit, param)
M (Fit, UDE)
E (Fit, UDE)
P (Fit, UDE)

D

E

F

Max. likelihood

$log(d_p)$

Profile (known)
Profile (param.)
Profile (UDE)
d (true)
95% conf. interval

G

H

I

$R(P;p)$

P

$R(P;\boldsymbol{p}) = 1.0\frac{0.4^4}{0.4^4 + P^4}$

$R(P;\boldsymbol{p}) = v\frac{K^n}{K^n + P^n}$

$R(P;\boldsymbol{p}) = U(P;\boldsymbol{\theta})$